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Substrate holding apparatus, substrate processing apparatus having substrate holding apparatus, and substrate processing method

Abstract

A substrate holding apparatus which can enhance the efficiency of processing of a substrate, such as a wafer, is disclosed. The substrate holding apparatus for holding a substrate and rotating the substrate, includes: rollers capable of contacting a periphery of the substrate; a roller rotating mechanism configured to rotate the rollers; and eccentric shafts coupling the rollers and the roller rotating mechanism, the eccentric shafts having first shaft portions and second shaft portions, the second shaft portions being eccentric relative to the first shaft portions, the first shaft portions being secured to the roller rotating mechanism, and the rollers being secured to the second shaft portions, respectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional of U.S. patent application Ser. No. 16/168,301 filed Oct. 23, 2018, which claims priority to Japanese Patent Application Number 2017-208327 filed Oct. 27, 2017, each of which are incorporated by reference in their entirety for any and all purposes.

BACKGROUND

(1) Devices such as memory circuits, logic circuits and image sensors (e.g. CMOS sensors) are becoming more highly integrated these days. In a process for forming such a device, foreign matters such as fine particles and dust may adhere to the device. Foreign matter adhering to a device can cause a short-circuit between interconnects or a circuit defect. Therefore, in order to enhance the reliability of the device, it is necessary to clean a wafer on which the device is formed to remove the foreign matter on the wafer.

(2) The above-described foreign matters, such as fine particles and dust, may adhere also to a back surface (bare-silicon surface) of a wafer. When such foreign matter adheres to the back surface of a wafer, the wafer may separate from a stage reference surface, or a front surface of the wafer may be inclined with respect to the stage reference surface in an exposure apparatus, resulting in patterning deviation or deviation of focal distance. To prevent such problems, it is necessary to remove foreign matter adhering to a back surface of a wafer.

(3) Japanese Laid-Open Patent Publication No. 2013-172019 discloses a substrate processing apparatus for processing a surface of a substrate by rubbing a scrubbing member against the

substrate surface. This substrate processing apparatus is configured to hold a periphery of a substrate with chucks, and to press the scrubbing member against a surface of the substrate while rotating the chucks around the center of the substrate.

SUMMARY OF THE INVENTION

(4) Recently, there is an increasing demand for an apparatus which can more efficiently process an entirety of a substrate surface. Therefore, there is provided a substrate holding apparatus which can enhance the efficiency of processing of a substrate such as a wafer. In addition, there are provided a substrate processing apparatus and a substrate processing method for processing a substrate surface with use of such a substrate holding apparatus.

(5) Embodiments, which will be described below, relate to a substrate holding apparatus for holding and rotating a substrate such as a wafer. The below-described embodiments also relate to a substrate processing apparatus and a substrate processing method for processing a surface of a substrate while rotating the substrate by the substrate holding apparatus.

(6) In an embodiment, there is provided a substrate holding apparatus for holding a substrate and rotating the substrate, comprising: rollers capable of contacting a periphery of the substrate; a roller rotating mechanism configured to rotate the rollers; and eccentric shafts coupling the rollers and the roller rotating mechanism, the eccentric shafts having first shaft portions and second shaft portions, the second shaft portions being eccentric relative to the first shaft portions, the first shaft portions being secured to the roller rotating mechanism, and the rollers being secured to the second shaft portions, respectively.

(7) In an embodiment, the roller rotating mechanism includes motors coupled to the first shaft portions, respectively, and an operation controller configured to rotate the motors at the same speed and in the same phase.

(8) In an embodiment, the substrate holding apparatus further comprises counter weights secured to the eccentric shafts, respectively, each of the counter weights and corresponding one of the rollers being arranged symmetrically with respect to corresponding one of the first shaft portions.

(9) In an embodiment, there is provided a substrate processing apparatus comprising: the substrate holding apparatus described above; and a processing head for bringing a processing tool into contact with a first surface of a substrate to process the first surface.

(10) In an embodiment, there is provided a substrate holding apparatus for holding a substrate and rotating the substrate, comprising: a drive roller capable of contacting a periphery of the substrate; a drive shaft to which the drive roller is secured; an orbital table rotatably supporting the drive shaft; a base plate located away from the orbital table; a first motor secured to the base plate; a first torque transmitting mechanism coupling the first motor and the drive shaft; and a translational rotating mechanism configured to cause the orbital table to make a translational rotating motion.

(11) In an embodiment, the substrate holding apparatus further comprises: a clamp roller capable of contacting the periphery of the substrate; an eccentric shaft rotatably supporting the clamp roller, the eccentric shaft being rotatably supported by the orbital table; a rotary actuator secured to the base plate; and a second torque transmitting mechanism coupling the rotary actuator and the eccentric shaft.

(12) In an embodiment, the substrate holding apparatus further comprises a counter weight secured to the eccentric shaft, wherein the eccentric shaft has a first shaft portion and a second shaft portion, the first shaft portion being coupled to the second torque transmitting mechanism, the second shaft portion being eccentric relative to the first shaft portion, the clamp roller is rotatably coupled to the second shaft portion, and the counter weight and the clamp roller are arranged symmetrically with respect to the first shaft portion.

(13) In an embodiment, the substrate holding apparatus further comprises an operation controller, wherein the translational rotating mechanism includes a main eccentric shaft rotatably coupled to the orbital table, a second motor coupled to the main eccentric shaft, and eccentric joints coupling the orbital table and the base plate, the second motor is secured to the base plate, and the operation

controller is configured to independently control a rotational speed of the first motor and a rotational speed of the second motor.

(14) In an embodiment, the translational rotating mechanism has counter weights secured to the eccentric joints, respectively, each of the eccentric joints includes a first joint shaft portion rotatably coupled to the base plate, and a second joint shaft portion rotatably coupled to the orbital table, the second joint shaft portion is eccentric relative to the first joint shaft portion, the second joint shaft portion and corresponding one of the counter weights are arranged symmetrically with respect to the first joint shaft portion, the main eccentric shaft includes a first main shaft portion coupled to the second motor, and a second main shaft portion rotatably coupled to the orbital table, and the second main shaft portion is eccentric relative to the first main shaft portion.

(15) In an embodiment, the first torque transmitting mechanism comprises a universal joint or a magnet gear.

(16) In an embodiment, the second torque transmitting mechanism comprises a universal joint or a magnet gear.

(17) In an embodiment, there is provided a substrate processing apparatus comprising: the substrate holding apparatus described above; and a processing head for bringing a processing tool into contact with a first surface of a substrate to process the first surface.

(18) In an embodiment, there is provided a substrate processing method comprising: bringing rollers into contact with a periphery of a substrate; rotating the rollers about their respective own axes while causing the rollers to make a circular motion, thereby rotating the substrate about an axis of the substrate and causing the substrate to make a circular motion; and bringing a processing tool into contact with the substrate, which is rotating and making the circular motion, to process the substrate.

(19) According to the above-described embodiments, the substrate holding apparatus can rotate a substrate, such as a wafer, about its axis while causing the substrate to make a circular motion. The combination of the circular motion of the substrate and the rotation of the substrate on its axis can increase the speed of the substrate at any point on the surface of the substrate. Therefore, when the processing head is pressed against the surface of the substrate, the relative speed between the processing head and the surface of the substrate can be increased, thereby increasing a processing rate of the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view schematically showing an embodiment of a substrate holding apparatus;

(2) FIG. 2 is a side view of the substrate holding apparatus shown in FIG. 1;

(3) FIG. 3 is a bottom view of the substrate holding apparatus shown in FIG. 1;

(4) FIG. 4 is a cross-sectional view taken along line A-A of FIG. 3;

(5) FIGS. 5A, 5B, 5C, 5D are schematic diagrams illustrating the operation of the substrate holding apparatus when receiving a wafer;

(6) FIG. 6 is a side view showing another embodiment of a substrate holding apparatus;

(7) FIG. 7 is a plan view of the substrate holding apparatus shown in FIG. 6;

(8) FIG. 8 is a bottom view of the substrate holding apparatus;

(9) FIG. 9 is a plan view illustrating a state of the substrate holding apparatus when clamp rollers are away from a periphery of a wafer;

(10) FIG. 10 is a cross-sectional view taken along line B-B of FIG. 6;

(11) FIG. 11 is an enlarged view of one of four eccentric joints;

(12) FIG. 12 is a side view showing yet another embodiment of a substrate holding apparatus;

(13) FIG. 13 is an enlarged view of a first magnet gear shown in FIG. 12;

(14) FIG. 14A is a diagram showing a drive magnet rotor, and FIG. 14B is a diagram showing a driven magnet rotor;

(15) FIG. 15 is a plan view schematically showing an embodiment of a substrate processing apparatus including the substrate holding apparatus described with reference to FIGS. 1 through 5; and

(16) FIG. 16 is a side view of the substrate processing apparatus shown in FIG. 15.

DESCRIPTION OF EMBODIMENTS

(17) Embodiments will now be described with reference to the drawings. FIG. 1 is a plan view schematically showing an embodiment of a substrate holding apparatus, FIG. 2 is a side view of the substrate holding apparatus shown in FIG. 1, and FIG. 3 is a bottom view of the substrate holding apparatus shown in FIG. 1. The substrate holding apparatus of this embodiment is configured to hold and rotate a wafer W which is an exemplary substrate. The substrate holding apparatus includes a plurality of rollers 11a and a plurality of rollers 11b, both of which are capable of contacting a periphery of the wafer W, a roller rotating mechanism 12 for rotating the rollers 11a, 11b, a plurality of eccentric shafts 13a coupling the rollers 11a and the roller rotating mechanism 12 to each other, and a plurality of eccentric shafts 13b coupling the rollers 11b and the roller rotating mechanism 12 to each other. Each roller 11a has a wafer holding surface (or a substrate holding surface) 31a for holding the periphery of the wafer W, and each roller 11b has a wafer holding surface (or a substrate holding surface) 31b for holding the periphery of the wafer W. Each roller 11a and each roller 11b have the same construction and the same size.

(18) The eccentric shafts 13a each have a first shaft portion 14a, and a second shaft portion 15a which is eccentric relative to the first shaft portion 14a. The first shaft portion 14a is secured to the roller rotating mechanism 12, and the roller 11a is secured to the second shaft portion 15a. The eccentric shafts 13b each have a first shaft portion 14b, and a second shaft portion 15b which is eccentric relative to the first shaft portion 14b. The first shaft portion 14b is secured to the roller rotating mechanism 12, and the roller 11b is secured to the second shaft portion 15b.

(19) The roller rotating mechanism 12 includes a plurality of motors 29a coupled respectively to the first shaft portions 14a of the eccentric shafts 13a, a plurality of motors 29b coupled respectively to the first shaft portions 14b of the eccentric shafts 13b, and an operation controller 40 for causing the motors 29a, 29b to rotate at the same speed and in the same phase. Each motor 29a is configured to rotate the eccentric shaft 13a about the first shaft portion 14a, and each motor 29b is configured to rotate the eccentric shaft 13b about the first shaft portion 14b. The motors 29a, 29b are coupled to the operation controller 40.

(20) The operations of the motors 29a, 29b are controlled by the operation controller 40. More specifically, the operation controller 40 starts and stops all the motors 29a, 29b with the same timing, and rotates all the motors 29a, 29b in the same direction. Furthermore, during operation of the motors 29a, 29b, the operation controller 40 synchronizes the rotational speeds and phases of the motors 29a, 29b. Accordingly, all the rollers 11a, 11b rotate in the same direction, at the same rotational speed and in the same phase and, in addition, make a circular motion.

(21) The substrate holding apparatus of this embodiment includes two rollers 11a, two rollers 11b, two eccentric shafts 13a, two eccentric shafts 13b, two motors 29a, and two motors 29b. It is noted, however, that the numbers of these components are not limited to those of this embodiment.

(22) The substrate holding apparatus further includes a base plate 20. The eccentric shafts 13a, 13b extend through the base plate 20. The roller rotating mechanism 12 is disposed at one side of the base plate 20, and the rollers 11a, 11b are disposed at the opposite side of the base plate 20.

(23) A plurality of linear motion guides 26 are secured to a lower surface of the base plate 20, and a movable base 21 is supported by the linear motion guides 26. The linear motion guides 26 are coupled to the movable base 21. The linear motion guides 26 are configured to restrict movement of the movable base 21 to a linear movement in a direction parallel to the lower surface of the base plate 20. An actuator 18 is further secured to the lower surface of the base plate 20. The movable

base **21** is coupled to the actuator **18**. The actuator **18** is disposed such that it moves the movable base **21** parallel to the base plate **20**. The operation of the actuator **18** is controlled by the operation controller **40**. The substrate holding apparatus further includes a support plate **25** which rotatably supports the two eccentric shafts **13b** via bearings **24**. The two motors **29b** are secured to the support plate **25** via motor supports **27b**.

(24) FIG. **4** is a cross-sectional view taken along line A-A of FIG. **3**. The substrate holding apparatus further includes a pivot shaft **23** secured to the support plate **25**. The support plate **25** is disposed below the movable base **21**, and the pivot shaft **23** is rotatably supported by a bearing **28**, which is fixed to the movable plate **21**. The movable base **21** and the support plate **25** are coupled together via the pivot shaft **23**, and the support plate **25** is pivotable relative to the movable plate **21**. The two eccentric shafts **13b** (see FIG. **2**), the support plate **25**, and the two motors **29b** (see FIG. **2**) are configured to be capable of pivoting together on the pivot shaft **23**. As shown by arrows in FIG. **2**, the actuator **18** moves the two eccentric shafts **13b**, the support plate **25**, the pivot shaft **23**, and the two motors **29b**, together with the movable base **21**, parallel to the base plate **20**.

(25) Referring back to FIG. **2**, bearings **19** are fixed to the base plate **20**. The two eccentric shafts **13a** are rotatably supported by the base plate **20** via the bearings **19**, respectively. The positions of the eccentric shafts **13a** are fixed. The two motors **29a** are secured to the base plate **20**. More specifically, two motor supports **27a** are secured to the lower surface of the base plate **20**, and the two motors **29a** are secured to the motor supports **27a**, respectively. Thus, the two motors **29a** are secured to the base plate **20** via the motor supports **27a**.

(26) According to the above-described construction, when the two eccentric shafts **13b** move in a direction closer to the two eccentric shafts **13a**, the wafer **W** is held by the two rollers **11a** and the two rollers **11b**. Since the two rollers **11b** are pivotable on the pivot shaft **23**, the positions of the rollers **11b** can be automatically adjusted when the four rollers **11a**, **11b** are holding the wafer **W**. The wafer **W** is released from the two rollers **11a** and the two rollers **11b** when the two eccentric shafts **13b** move in a direction away from the two eccentric shafts **13a**. Though the two eccentric shafts **13a** and the two eccentric shafts **13b**, arranged around the axis **CP** of the substrate holding apparatus, are provided in this embodiment, the numbers of these components are not limited to those of this embodiment. For example, it is possible to provide one eccentric shaft **13a** and two eccentric shafts **13b**, arranged around the axis **CP** at equal intervals, i.e. 120-degree intervals.

(27) The second shaft portion **15a** of each eccentric shaft **13a** is decentered by a distance e from the first shaft portion **14a**. Accordingly, when the motor **29a** is in motion, the roller **11a**, while rotating about the second shaft portion **15a** of the eccentric shaft **13a**, makes a circular motion with a radius e . The axis of the roller **11a** coincides with the axis of the second shaft portion **15a**. Therefore, the roller **11a**, while rotating about its axis, makes a circular motion with a radius e around the axis of the first shaft portion **14a**.

(28) Similarly, the second shaft portion **15b** of each eccentric shaft **13b** is decentered by a distance e from the first shaft portion **14b**. Accordingly, when the motor **29b** is in motion, the roller **11b**, while rotating about the second shaft portion **15b** of the eccentric shaft **13b**, makes a circular motion with a radius e . The axis of the roller **11b** coincides with the axis of the second shaft portion **15b**. Therefore, the roller **11b**, while rotating about its axis, makes a circular motion with a radius e around the axis of the first shaft portion **14b**. The circular motion is defined herein as a movement of an object in a circular orbit.

(29) As described above, the operation controller **40** rotates all the motors **29a**, **29b** at the same speed and in the same phase. Therefore, when the operation controller **40** sets the motors **29a**, **29b** in motion, the wafer **W**, held by the rollers **11a**, **11b**, makes a circular motion with a radius e , while rotating about the axis of the wafer **W**.

(30) In one embodiment, the roller rotating mechanism **12** may include a plurality of driven pulleys (not shown) secured to the first shaft portions **14a**, **14b** of the eccentric shafts **13a**, **13b**, at least one motor (not shown), a drive pulley(s) (not shown) secured to the rotating shaft(s) of the at least one

motor, and at least one belt (not shown) riding on the driven pulleys and the drive pulley(s). In this case, all the rollers **11a**, **11b** rotate at the same speed and in the same phase by the motor(s) to which the drive pulley(s) is secured. Thus, also in this case, when the motor(s), to which the drive pulley(s) is secured, is in motion, the wafer **W** held by the rollers **11a**, **11b** makes a circular motion with a radius e while the wafer **W** is rotating about the axis of the wafer **W**.

(31) A counter weight **17a** and a counter weight **17b** are secured to each eccentric shaft **13a** and each eccentric shaft **13b**, respectively. More specifically, the counter weights **17a**, **17b** are respectively secured to intermediate shaft portions **16a**, **16b** that couple the first shaft portions **14a**, **14b** and the second shaft portions **15a**, **15b**, respectively. The counter weight **17a** and the roller **11a** are arranged symmetrically with respect to the first shaft portion **14a**. The weight of the counter weight **17a** is such that a centrifugal force, which is generated in the radial direction from the first shaft portion **14a** toward the roller **11a** when the eccentric shaft **13a** is rotating about the first shaft portion **14a**, is canceled by a centrifugal force acting on the counter weight **17a**.

(32) Similarly, the counter weight **17b** and the roller **11b** are arranged symmetrically with respect to the first shaft portion **14b**. The weight of the counter weight **17b** is such that a centrifugal force, generated in the radial direction from the first shaft portion **14b** toward the roller **11b** when the eccentric shaft **13b** is rotating about the first shaft portion **14b**, is canceled by a centrifugal force acting on the counter weight **17b**. Such counter weights **17a**, **17b** can prevent vibration of the eccentric shafts **13a**, **13b** due to a weight imbalance during rotation of the eccentric shafts **13a**, **13b**.

(33) FIGS. 5A through 5D are schematic diagrams illustrating the operation of the substrate holding apparatus when receiving the wafer **W**. As shown in FIG. 5A, before the substrate holding apparatus receives the wafer **W**, the actuator **18** (see FIG. 2) is set in motion to move the rollers **11b** in a direction away from the rollers **11a**. The rollers **11a** and the rollers **11b** are decentered outward. As shown in FIG. 5B, the wafer **W** is transported to the substrate holding apparatus by a not-shown transport device. Thereafter, as shown in FIG. 5C, while the wafer **W** is located between the rollers **11a** and the rollers **11b**, the eccentric shafts **13a** are rotated 180 degrees to decenter the rollers **11a** inward. Thereafter, the actuator **18** moves the rollers **11b** in a direction closer to the rollers **11a** until the rollers **11b** come into contact with the wafer **W**. The periphery of the wafer **W** is held by the wafer holding surfaces **31a** of the rollers **11a** and the wafer holding surfaces **31b** of the rollers **11b**. When the wafer **W** is to be removed from the substrate holding apparatus, the process steps shown in FIGS. 5A through 5D are performed in the opposite order.

(34) The operation of the substrate holding apparatus, holding the wafer **W**, is as follows. The operation controller **40** instructs the motors **29a**, **29b** to rotate the eccentric shafts **13a**, **13b** in the same direction about the first shaft portions **14a**, **14b**. The operation controller **40** starts the motors **29a**, **29b** with the same timing. The operation controller **40** synchronizes the rotational speeds and the phases of all the motors **29a**, **29b**. The above operation allows the wafer **W** to rotate about its axis while causing the wafer **W** to perform a circular motion with a radius e .

(35) The substrate holding apparatus according to this embodiment has a simple construction and yet can rotate the wafer **W** about its axis while allowing the wafer **W** to make a circular motion. The combination of the circular motion of the wafer **W** and the rotation of the wafer **W** about the axis of the wafer **W** can increase the speed of the wafer **W** at any point on the surface of the wafer **W**. Therefore, when the below-described processing head is pressed against the surface of the wafer **W**, the relative speed between the processing head and the surface of the wafer **W** can be increased, thereby increasing a processing rate of the wafer **W**.

(36) Another embodiment will now be described. FIG. 6 is a side view showing another embodiment of a substrate holding apparatus, and FIG. 7 is a plan view of the substrate holding apparatus shown in FIG. 6. The substrate holding apparatus of this embodiment includes two drive rollers **61** which are capable of contacting a periphery of a wafer **W**, two drive shafts **64** to which the two drive rollers **61** are secured respectively, an orbital table **66** which rotatably supports the two drive shafts **64**, a base plate **68** arranged away from the orbital table **66**, two first motors **71**

secured to the base plate **68**, and two first universal joints **72** as first torque transmitting mechanisms coupling the two first motors **71** and the two drive shafts **64**, respectively. It is to be noted that FIG. **6** only shows one drive roller **61**, one drive shaft **64**, one first motor **71**, and one first universal joint **72**.

(37) Though the substrate holding apparatus of this embodiment includes two drive rollers **61**, two drive shafts **64**, two first motors **71**, and two first universal joints **72**, the numbers of these components are not limited to those of this embodiment.

(38) Each drive roller **61** has a wafer holding surface (or a substrate holding surface) **61a** for holding the periphery of the wafer W. Each drive roller **61** is secured to one end of each drive shaft **64** so that the drive roller **61** and the drive shaft **64** can rotate together about the axis of the drive roller **61**. The drive shaft **64** extends perpendicularly to the orbital table **66**. The orbital table **66** includes two bearings **75** (only one bearing **75** is shown in FIG. **6**) that rotatably support the two drive shafts **64**, and a pedestal **82** holding these bearings **75**. The pedestal **82** of this embodiment has a plate-like shape, while there is no particular limitation on the shape of the pedestal **82** as long as a sufficient mechanical rigidity is ensured. The drive shafts **64** extend through the orbital table **66**, and the other ends of the drive shafts **64** are coupled to one ends of the first universal joints **72**, respectively.

(39) In this embodiment, the base plate **68** is located below the orbital table **66**. The two first universal joints **72** are arranged between the orbital table **66** and the base plate **68**. Each of the first universal joints **72** is an articulated universal joint having a plurality of joints. One end of each first universal joint **72** is coupled to the drive shaft **64**, while the other end of the first universal joint **72** is coupled to the shaft **71a** of the first motor **71**.

(40) The first motors **71** are secured to the base plate **68**. More specifically, first motor platforms **85** are secured to the lower surface of the base plate **68**, and the first motors **71** are secured to the first motor platforms **85**, respectively. Thus, the first motors **71** are secured to the base plate **68** via the first motor platforms **85**. The shafts **71a** of the first motors **71** are rotatably supported by bearings **86** fixed to the base plate **68**. Each shaft **71a** may be comprised of a plurality of shafts coupled in tandem by a coupling(s).

(41) When the first motor **71** is in operation, the rotation of the shaft **71a** of the first motor **71** is transmitted through the first universal joint **72** to the drive shaft **64**, and the drive shaft **64** and the drive roller **61** rotate together about the axis of the drive roller **61**.

(42) The substrate holding apparatus further includes two clamp rollers **90** which are capable of contacting the periphery of the wafer W, two eccentric shafts **93** which rotatably support the two clamp rollers **90**, two rotary actuators **96** secured to the base plate **68**, and two second universal joints **98** as second torque transmitting mechanisms coupling the two rotary actuators **96** and the two eccentric shafts **93**, respectively. It is to be noted that FIG. **6** only shows one clamp roller **90**, one eccentric shaft **93**, one rotary actuator **96**, and one second universal joint **98**.

(43) Though the substrate holding apparatus of this embodiment includes two clamp rollers **90**, two eccentric shafts **93**, two rotary actuators **96**, and two second universal joints **98**, the numbers of these components are not limited to those of this embodiment. Though the two drive shafts **64** and the two eccentric shafts **93**, arranged around the axis CP of the substrate holding apparatus, are provided in this embodiment, the numbers of these components are not limited to those of this embodiment.

(44) Each clamp roller **90** has a wafer holding surface (or a substrate holding surface) **90a** for holding the periphery of the wafer W. Each eccentric shaft **93** has a first shaft portion **93a** coupled to the second universal joint **98**, and a second shaft portion **93b** which is eccentric relative to the first shaft portion **93a**. A counter weight **100** is secured to each of the eccentric shafts **93**. The counter weight **100** and the clamp roller **90** are arranged symmetrically with respect to the first shaft portion **93a**. The weight of the counter weight **100** is such that a centrifugal force, which acts on the second shaft portion **93b** (or the clamp roller **90**) by the below-described translational

rotating motion, is canceled by a centrifugal force acting on the counter weight **100**. The counter weight **100** is provided to stabilize a clamping force of the clamp roller **90** applied to the wafer **W** during the below-described translational rotating motion, i.e. to prevent a clamping force of the clamp roller **90** applied to the wafer **W** from fluctuating due to the translational rotating motion.

(45) The clamp rollers **90** are rotatably coupled to the second shaft portions **93b**, respectively. More specifically, each of the second shaft portions **93b** has a bearing **101** disposed therein, and each clamp roller **90** is rotatably supported by the bearing **101**. Such construction enables the clamp rollers **90** to rotate relative to the eccentric shafts **93**.

(46) The eccentric shafts **93** are rotatably supported by the orbital table **66**. More specifically, the orbital table **66** includes two bearings **104** (only one bearing **104** is shown in FIG. **6**) held by the pedestal **82**. The first shaft portions **93a** of the two eccentric shafts **93** are rotatably supported by the bearings **104**, respectively. The two second universal joints **98** are located between the orbital table **66** and the base plate **68**. Each second universal joint **98** is an articulated universal joint having a plurality of joints. One ends of the second universal joints **98** are coupled to the eccentric shafts **93**, while the other ends of the second universal joints **98** are coupled to the rotary actuators **96** secured to the base plate **68**.

(47) Each of the rotary actuators **96** includes an air cylinder **110**, a crank **111** coupled to the air cylinder **110**, and a rotating shaft **112** secured to the crank **111**. The air cylinder **110** is rotatably secured via a pivot shaft **115** to the lower surface of the base plate **68**. The pivot shaft **115** extends perpendicularly to the base plate **68**, and the air cylinder **110** can rotate about the pivot shaft **115** in a direction parallel to the base plate **68**.

(48) The rotating shaft **112** is rotatably supported by a bearing **117** fixed to the base plate **68**. One end of the rotating shaft **112** is secured to the crank **111**, while the other end of the rotating shaft **112** is secured to the second universal joint **98**. The rotating shaft **112** may be comprised of a plurality of shafts coupled in tandem by a coupling(s).

(49) FIG. **8** is a bottom view of the substrate holding apparatus. A piston rod **110a** of the air cylinder **110** is rotatably coupled to an end portion of the crank **111**. The crank **111** converts a linear motion of the piston rod **110a** of the air cylinder **110** into a rotating motion. This rotating motion of the crank **111** rotates the rotating shaft **112**.

(50) When the rotary actuator **96** is in motion, the second universal joint **98** (see FIG. **6**) rotates by a certain angle. This rotation of the second universal joint **98** rotates the eccentric shaft **93** (see FIG. **6**) and, at the same time, causes the clamp roller **90** (see FIG. **6**) to make a circular motion. This circular motion of the clamp roller **90** is an arc-like movement of the clamp roller **90** around the axis of the first shaft portion **93a** (see FIG. **6**) of the eccentric shaft **93**. In particular, when the rotary actuator **96** rotates the eccentric shaft **93** in one direction, the clamp roller **90** moves in a direction closer to the periphery of the wafer **W**. When the rotary actuator **96** rotates the eccentric shaft **93** in the opposite direction, the clamp roller **90** moves in a direction away from the periphery of the wafer **W**.

(51) FIG. **7** illustrates a state of the substrate holding apparatus when the clamp rollers **90** have moved in a direction closer to the periphery of the wafer **W**, until the wafer holding surfaces **90a** of the clamp rollers **90** are brought into contact with the periphery of the wafer **W**. As shown in FIG. **7**, when the wafer holding surfaces **90a** of the clamp rollers **90** are brought into contact with the periphery of the wafer **W**, the wafer **W** is held by the two drive rollers **61** and the two clamp rollers **90**. When the drive rollers **61** rotate while the periphery of the wafer **W** is in contact with the drive rollers **61** and the clamp rollers **90**, the rotations of the drive rollers **61** are transmitted to the wafer **W**, whereby the wafer **W** is rotated about the axis of the wafer **W**.

(52) The positions of the clamp rollers **90**, shown in FIG. **7**, are clamping positions where the clamp rollers **90** hold the wafer **W**. When the rotary actuators **96** (see FIGS. **6** and **8**) rotate the eccentric shafts **93** in the opposite direction, the clamp rollers **90** move away from the periphery of the wafer **W**. FIG. **9** illustrates a state of the substrate holding apparatus when the clamp rollers **90**

are away from the periphery of the wafer W. The positions of the clamp rollers **90**, shown in FIG. **9**, are unclamping positions where the clamp rollers **90** release the wafer W.

(53) As shown in FIG. **6**, the substrate holding apparatus further includes a translational rotating mechanism **120** for causing the orbital table **66** to make a translational rotating motion. The translational rotating motion of the orbital table **66** is a circular motion of the orbital table **66** in a circular orbit without rotation of the orbital table **66** itself about the axis of the orbital table **66**. This translational rotating motion is an example of a circular motion. The translational rotating mechanism **120** includes a main eccentric shaft **121** rotatably coupled to the orbital table **66**, a second motor **124** coupled to the main eccentric shaft **121**, and a plurality of eccentric joints **128** coupling the orbital table **66** and the base plate **68**.

(54) The main eccentric shaft **121** includes a first main shaft portion **121a** coupled to the second motor **124**, and a second main shaft portion **121b** rotatably coupled to the orbital table **66**. The second main shaft portion **121b** is decentered by a distance e from the first main shaft portion **121a**. A bearing **125** is fixed to the base plate **68**, and the first main shaft portion **121a** is rotatably supported by the bearing **125**. The orbital table **66** has a bearing **127** fixed to the pedestal **82**, and the second main shaft portion **121b** is rotatably supported by the bearing **127**.

(55) The second motor **124** is secured to the base plate **68**. More specifically, a second motor platform **129** is secured to the lower surface of the base plate **68**, and the second motor **124** is secured to the second motor platform **129**. Thus, the second motor **124** is secured to the base plate **68** via the second motor platform **129**. When the second motor **124** is in motion, the main eccentric shaft **121** is rotated to cause the orbital table **66**, coupled to the second main shaft portion **121b** of the main eccentric shaft **121**, to make a translational rotating motion. In one embodiment, the second motor **124** and the main eccentric shaft **121** may be coupled via a rotation transmitting mechanism (not shown) comprised of, for example, a combination of a belt and pulleys.

(56) When the second motor **124** rotates the main eccentric shaft **121**, the orbital table **66** makes a translational rotating motion. Since the drive shafts **64** and the eccentric shafts **93** are supported by the orbital table **66**, the drive rollers **61**, the clamp rollers **90**, the drive shafts **64**, the eccentric shafts **93**, and the orbital table **66** make a translational rotating motion together. With the translational rotating motion of the drive shafts **64** and the eccentric shafts **93**, the wafer W held by the drive rollers **61** and the clamp rollers **90** also makes a translational rotating motion.

(57) The substrate holding apparatus further includes an operation controller **134**. The first motors **71** and the second motor **124** are electrically connected to the operation controller **134**. The operation controller **134** is configured to be capable of controlling the operation of the first motors **71** and the operation of the second motor **124** independently. More specifically, the operation controller **134** can independently control the start and stop of the first motors **71**, the start and stop of the second motor **124**, the rotational speed and the rotational direction of the first motors **71**, and the rotational speed and the rotational direction of the second motor **124**. For example, the operation controller **134** can allow the first motors **71** and the second motor **124** to rotate at the same rotational speed or different rotational speeds. Further, the operation controller **134** can allow the first motors **71** and the second motor **124** to rotate in the same rotational direction or different rotational directions. The above-described operation of the rotary actuators **96** is also controlled by the operation controller **134**.

(58) The eccentric joints **128** function to keep the orbital table **66** parallel to the base plate **68** while permitting the translational rotating motion of the orbital table **66**. While only two eccentric joints **128** are depicted in FIG. **6**, at least two eccentric joints **128** are disposed around the main eccentric shaft **121**. In this embodiment, four eccentric joints **128** are disposed around the main eccentric shaft **121**, as shown in FIG. **10**. FIG. **10** is a cross-sectional view taken along line B-B of FIG. **6**.

(59) FIG. **11** is an enlarged view of one of the four eccentric joints **128**. The other eccentric joints **128** have the same construction, and therefore a duplicate description thereof will be omitted. As shown in FIG. **11**, the eccentric joint **128** includes a first joint shaft portion **128a** rotatably coupled

to the base plate **68**, and a second joint shaft portion **128b** rotatably coupled to the orbital table **66**. A bearing **135** is fixed to the base plate **68**, and the first joint shaft portion **128a** is rotatably supported by the bearing **135**. The orbital table **66** has a bearing **136** fixed to the pedestal **82**, and the second joint shaft portion **128b** is rotatably supported by the bearing **136**.

(60) The second joint shaft portion **128b** is decentered by a distance e from the first joint shaft portion **128a**. This distance e is equal to the distance e by which the second main shaft portion **121b** of the main eccentric shaft **121** is eccentric relative to the first main shaft portion **121a**. Accordingly, when the orbital table **66** is making a translational rotating motion, the second joint shaft portion **128b** of the eccentric joint **128** rotates in a circular orbit with the same radius as that of the circular orbit in which the second main shaft portion **121b** of the main eccentric shaft **121** rotates. The thus-constructed eccentric joint **128** can support the orbital table **66** while allowing the orbital table **66** to make a translational rotating motion.

(61) A counter weight **140** is secured to the eccentric joint **128**. The counter weight **140** is secured to an intermediate shaft portion **128c** coupling the first joint shaft portion **128a** and the second joint shaft portion **128b**. The counter weight **140** and the second joint shaft portion **128b** are arranged symmetrically with respect to the first joint shaft portion **128a**. The weight of the counter weight **140** is equal to an equally-divided amount of the sum of the weight of the orbital table **66** and the weight of structures, located above the orbital table **66**, of the substrate holding apparatus. Specifically the weight of the counter weight **140** is such that it can cancel a centrifugal force generated in the above structures, including the orbital table **66**, by the translational rotating motion. Such counter weight **140** can prevent vibration of the above structures, including the orbital table **66**, and vibration of the wafer **W** due to a weight imbalance during rotation of the eccentric joint **128**.

(62) The operation of the thus-constructed substrate holding apparatus is as follows. The two rotary actuators **96** move the clamp rollers **90** to the unclamping positions shown in FIG. **9**. The wafer **W** is transported to the substrate holding apparatus by a not-shown transport device. While the wafer **W** is located between the drive rollers **61** and the clamp rollers **90**, the rotary actuators **96** move the clamp rollers **90** to the clamping positions shown in FIG. **7**. The periphery of the wafer **W** is held by the wafer holding surfaces **61a** of the drive rollers **61** and the wafer holding surfaces **90a** of the clamp rollers **90**.

(63) The first motors **71** rotate the drive rollers **61** about their own axes via the first universal joints **72** and the drive shafts **64**. The rotations of the drive rollers **61** are transmitted to the wafer **W**, whereby the wafer **W** rotates about the axis of the wafer **W**. The clamp rollers **90** are freely rotatable. Therefore, the clamp rollers **90** are rotated by the rotation of the wafer **W**. The second motor **124** of the translational rotating mechanism **120** rotates the main eccentric shaft **121**, thereby causing the orbital table **66** to make a translational rotating motion. The wafer **W**, the drive rollers **61**, the clamp rollers **90**, the drive shafts **64**, and the eccentric shafts **93** make a translational rotating motion together with the orbital table **66**.

(64) As shown in FIG. **6**, the drive shafts **64** are coupled to the first motors **71** via the first universal joints **72**, and the eccentric shafts **93** are coupled to the rotary actuators **96** via the second universal joints **98**. Therefore, the drive shafts **64** and the eccentric shafts **93** can make a translational rotating motion together with the orbital table **66**, while the positions of the first motors **71** and the positions of the rotary actuators **96** are fixed. The first motors **71** and the rotary actuators **96**, each of which have relatively heavy weight, are secured to the base plate **68**, and do not make a translational rotating motion. This configuration can stabilize the translational rotating motion of the orbital table **66**, and can reduce the electrical power necessary for the second motor **124** to cause the orbital table **66** to make the translational rotating motion. Further, the first motors **71** and the rotary actuators **96** are located below the base plate **68**. This arrangement eliminates the need for wiring and pipe connection to those portions which make a translational rotating motion, thus making it possible to avoid the risk of breaking of wire due to long-term operation.

(65) The rotation of the drive rollers **61** and the translational rotating motion of the orbital table **66** may be started simultaneously, or the rotation of the drive rollers **61** may be started before or after the start of the translational rotating motion of the orbital table **66**. The rotational speed of the drive rollers **61** may be the same as or different from the rotational speed of the translational rotating motion of the orbital table **66**. The start and stop of the rotation of the drive rollers **61**, the start and stop of the translational rotating motion of the orbital table **66**, the rotational speed of the drive rollers **61**, and the rotational speed of the translational rotating motion of the orbital table **66** are controlled by the operation controller **134**.

(66) The substrate holding apparatus according to this embodiment can rotate the wafer **W** about its own axis while causing the wafer **W** to make a translational rotating motion. The combination of the translational rotating motion of the wafer **W** and the rotation of the wafer **W** about its axis can increase the speed of the wafer **W** at any point on the surface of the wafer **W**. Therefore, when the below-described processing head is pressed against the surface of the wafer **W**, the relative speed between the processing head and the surface of the wafer **W** can be increased, thereby increasing a processing rate of the wafer **W**.

(67) FIG. **12** is a side view showing yet another embodiment of a substrate holding apparatus. The construction and operation of this embodiment, not particularly described here, are the same as those of the embodiment described above with reference to FIGS. **6** through **11**, and therefore a duplicate description thereof will be omitted.

(68) The substrate holding apparatus according to the embodiment shown in FIG. **12** includes two first magnet gears **151** as first torque transmitting mechanisms, and two second magnet gears **152** as second torque transmitting mechanisms. The first magnet gears **151** and the second magnet gears **152** are located between the orbital table **66** and the base plate **68**. One ends of the first magnet gears **151** are coupled to the drive shafts **64**, while the other ends of the first magnet gears **151** are coupled to the shafts **71a** of the first motors **71**. One ends of the second magnet gears **152** are coupled to the first shaft portions **93a** of the eccentric shafts **93**, while the other ends of the second magnet gears **152** are coupled to the rotating shafts **112** of the rotary actuators **96**.

(69) Each first magnet gear **151** and each second magnet gear **152** have the same construction and the same size. Therefore, the construction of the first magnet gear **151** will be described below, and a description of the second magnet gear **152** will be omitted. FIG. **13** is an enlarged view of the first magnet gear **151**. As shown in FIG. **13**, the first magnet gear **151** includes a drive magnet rotor **156**, a driven magnet rotor **157**, a drive coupling shaft **160** secured to the center of the drive magnet rotor **156**, and a driven coupling shaft **161** secured to the center of the driven magnet rotor **157**. The drive magnet rotor **156** and the driven magnet rotor **157** face each other with a slight gap between them. A diameter of the drive magnet rotor **156** is larger than a diameter of the driven magnet rotor **157**. The driven magnet rotor **157** is decentered by a distance e from the drive magnet rotor **156**. The drive coupling shaft **160** is coupled to the shaft **71a** of the first motor **71**, while the driven coupling shaft **161** is coupled to the drive shaft **64**.

(70) FIG. **14A** is a diagram showing the drive magnet rotor **156**, and FIG. **14B** is a diagram showing the driven magnet rotor **157**. The drive magnet rotor **156** has a plurality of permanent magnets **164** arranged in a circle. Similarly, the driven magnet rotor **157** has a plurality of permanent magnets **165** arranged in a circle. The torque of the drive magnet rotor **156** is transmitted to the driven magnet rotor **157** by a magnetic force generated between the permanent magnets **164** of the drive magnet rotor **156** and the permanent magnets **165** of the driven magnet rotor **157**.

(71) When the driven magnet rotor **157** is making a translational rotating motion together with the drive shaft **64**, the driven magnet rotor **157**, while keeping facing the drive magnet rotor **156**, makes a circular motion in a circumferential direction of the drive magnet rotor **156**. Therefore, the torque of the drive magnet rotor **156** can be transmitted to the driven magnet rotor **157**.

(72) FIG. **15** is a plan view schematically showing an embodiment of a substrate processing

apparatus including the substrate holding apparatus described above with reference to FIGS. **1** through **5**, and FIG. **16** is a side view of the substrate processing apparatus shown in FIG. **15**. The construction and operation of this embodiment, not particularly described here, are the same as those of the embodiment described above with reference to FIGS. **1** through **5**, and therefore a duplicate description thereof will be omitted. The substrate processing apparatus of this embodiment includes a substrate holding apparatus **10**, and a processing head **200** for bringing a processing tool **201** into contact with a first surface **1** of a wafer W, held by the substrate holding apparatus **10**, to process the first surface **1**. The processing head **200** is located below the wafer W held by the substrate holding apparatus **10**, and the position of the processing head **200** is fixed.

(73) In this embodiment, the first surface **1** of the wafer W is a back surface of the wafer W, i.e. a non-device surface where no device is formed or no device is to be formed. A second surface **2** of the wafer W, which is opposite from the first surface **1**, is a front surface, i.e. a device surface where a device(s) is formed or a device(s) is to be formed. In this embodiment, the wafer W, with its first surface **1** facing downward, is horizontally held by the substrate holding apparatus **10**.

(74) Detailed operation of the substrate processing apparatus of this embodiment is as follows. The substrate holding apparatus **10** brings the plurality of rollers **11a** and the plurality of rollers **11b** into contact with the periphery of the wafer W, and rotates the rollers **11a**, **11b** about their respective own axes and, in addition, causes the rollers **11a**, **11b** to make a circular motion, thereby rotating the wafer W about its axis and causing the wafer W to make a circular motion. While the wafer W is making the rotation and the circular motion, the processing head **200** places the processing tool **201** in contact with the first surface **1** of the wafer W to process the first surface **1**.

(75) In this embodiment, the processing tool **201** is longer than the radius of the wafer W. One end of the processing tool **201** protrudes outside the periphery of the wafer W, while the other end of the processing tool **201** extends across the axis CP of the substrate holding apparatus **10**.

Accordingly, the processing head **200** can bring the processing tool **201** into contact with the entirety of the first surface **1** of the rotating wafer W. This configuration enables the processing tool **201** to process the entirety of the first surface **1**, including the outermost area, of the wafer W. The processing head **200** is located in such a position as not to contact the rollers **11a**, **11b** and the eccentric shafts **13a**, **13b** when the wafer W is making a circular motion.

(76) The substrate holding apparatus according to this embodiment has a simple construction and yet can rotate the wafer W about its axis while allowing the wafer W to make a circular motion. The combination of the circular motion of the wafer W and the rotation of the wafer W about its axis can increase the speed of the wafer W at any point on the surface of the wafer W. Therefore, the relative speed between the processing tool **201** and the surface of the wafer W can be increased, thereby increasing a processing rate of the wafer W.

(77) In one embodiment, the processing tool **201** may be a polishing tool for polishing the wafer W. Examples of the polishing tool include a polishing tape and a grindstone. In another embodiment, the processing tool **201** may be a cleaning tool for cleaning the wafer W. A cleaning tape is an example of the cleaning tool. A tape made of nonwoven fabric is an example of the cleaning tape.

(78) In one embodiment, the substrate processing apparatus may include the substrate holding apparatus according to one of the embodiments described above with reference to FIGS. **6** through **14**, instead of the substrate holding apparatus described above with reference to FIGS. **1** through **5**. The construction and operation of the substrate holding apparatus **10** are the same as those of one of the embodiments described above with reference to FIGS. **6** through **14**, and therefore a duplicate description thereof is omitted.

(79) The previous description of embodiments is provided to enable a person skilled in the art to make and use the present invention. Moreover, various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles and specific examples defined herein may be applied to other embodiments. Therefore, the present invention is not intended to be

limited to the embodiments described herein but is to be accorded the widest scope as defined by limitation of the claims.

Claims

1. A substrate holding apparatus for holding a substrate and rotating the substrate, comprising: a drive roller capable of contacting a periphery of the substrate; a drive shaft to which the drive roller is secured; an orbital table rotatably supporting the drive shaft; a base plate located away from the orbital table; a first motor secured to the base plate; a first torque transmitting mechanism coupling the first motor and the drive shaft; and a translational rotating mechanism configured to cause the orbital table to make a translational rotating motion.
 2. The substrate holding apparatus according to claim 1, further comprising: a clamp roller capable of contacting the periphery of the substrate; an eccentric shaft rotatably supporting the clamp roller, the eccentric shaft being rotatably supported by the orbital table; a rotary actuator secured to the base plate; and a second torque transmitting mechanism coupling the rotary actuator and the eccentric shaft.
 3. The substrate holding apparatus according to claim 2, further comprising a counter weight secured to the eccentric shaft, wherein the eccentric shaft has a first shaft portion and a second shaft portion, the first shaft portion being coupled to the second torque transmitting mechanism, the second shaft portion being eccentric relative to the first shaft portion, and the second shaft portion being configured to be rotatable 360 degrees around the first shaft portion, the clamp roller is rotatably coupled to the second shaft portion, and the counter weight and the clamp roller are arranged symmetrically with respect to the first shaft portion.
 4. The substrate holding apparatus according to claim 2, wherein the second torque transmitting mechanism comprises a universal joint or a magnet gear.
 5. The substrate holding apparatus according to claim 1, further comprising an operation controller, wherein the translational rotating mechanism includes a main eccentric shaft rotatably coupled to the orbital table, a second motor coupled to the main eccentric shaft, and eccentric joints coupling the orbital table and the base plate, the second motor is secured to the base plate, and the operation controller is configured to independently control a rotational speed of the first motor and a rotational speed of the second motor.
 6. The substrate holding apparatus according to claim 5, wherein: the translational rotating mechanism has counter weights secured to the eccentric joints, respectively; each of the eccentric joints includes a first joint shaft portion rotatably coupled to the base plate, and a second joint shaft portion rotatably coupled to the orbital table; the second joint shaft portion is eccentric relative to the first joint shaft portion; the second joint shaft portion and corresponding one of the counter weights are arranged symmetrically with respect to the first joint shaft portion; the main eccentric shaft includes a first main shaft portion coupled to the second motor, and a second main shaft portion rotatably coupled to the orbital table; and the second main shaft portion is eccentric relative to the first main shaft portion.
 7. The substrate holding apparatus according to claim 1, wherein the first torque transmitting mechanism comprises a universal joint or a magnet gear.
 8. A substrate processing apparatus comprising: the substrate holding apparatus according to claim 1; and a processing head for bringing a processing tool into contact with a first surface of a substrate to process the first surface.
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